

STAKEHOLDER PACK

IMMUNE 永久研究網絡

Permanent research star map for the IMMUNE 3D immune-cell strategy game. Players spend 抗原樣本, 一度原質, 融合核心 and 生物質 to unlock 200 real nodes: identities, mobility, fusion, status chemistry, and Apex.

200 Research nodes	31 Character identities	6→10 Protocol bandwidth	5/200 Demo completed
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PRODUCT SUMMARY

The live entry is [ui/immune-research-network/index.html](#). Space = family sector. Shape/icon = research class. Ring + symbol = unlock state. Demo reveals 22 nodes and completes CORE-IMMUNE, T/B identity, BASE-T-01 and BASE-T-02. Selected start node: BASE-T-03 T 細胞固定炮台專精.

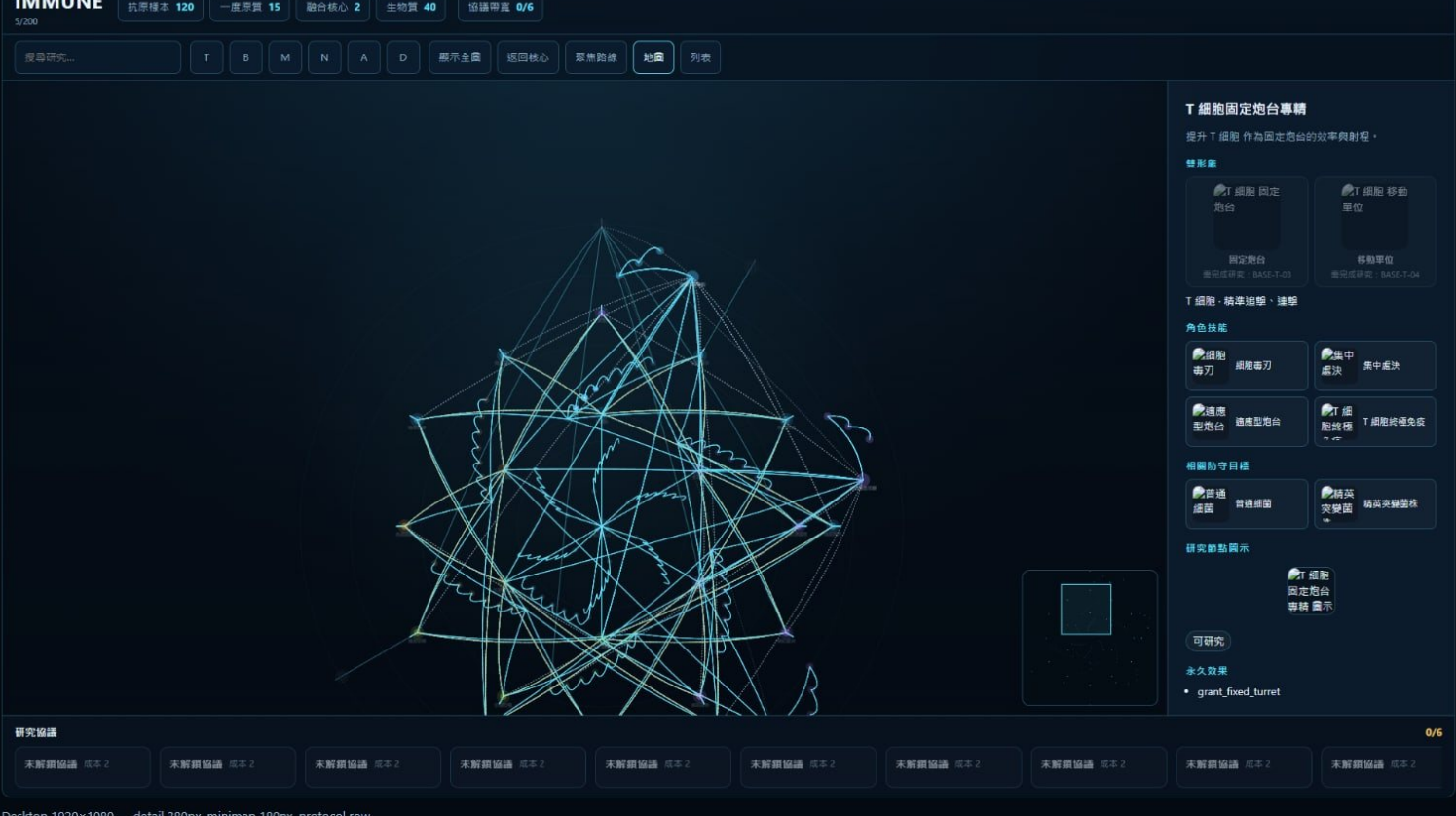
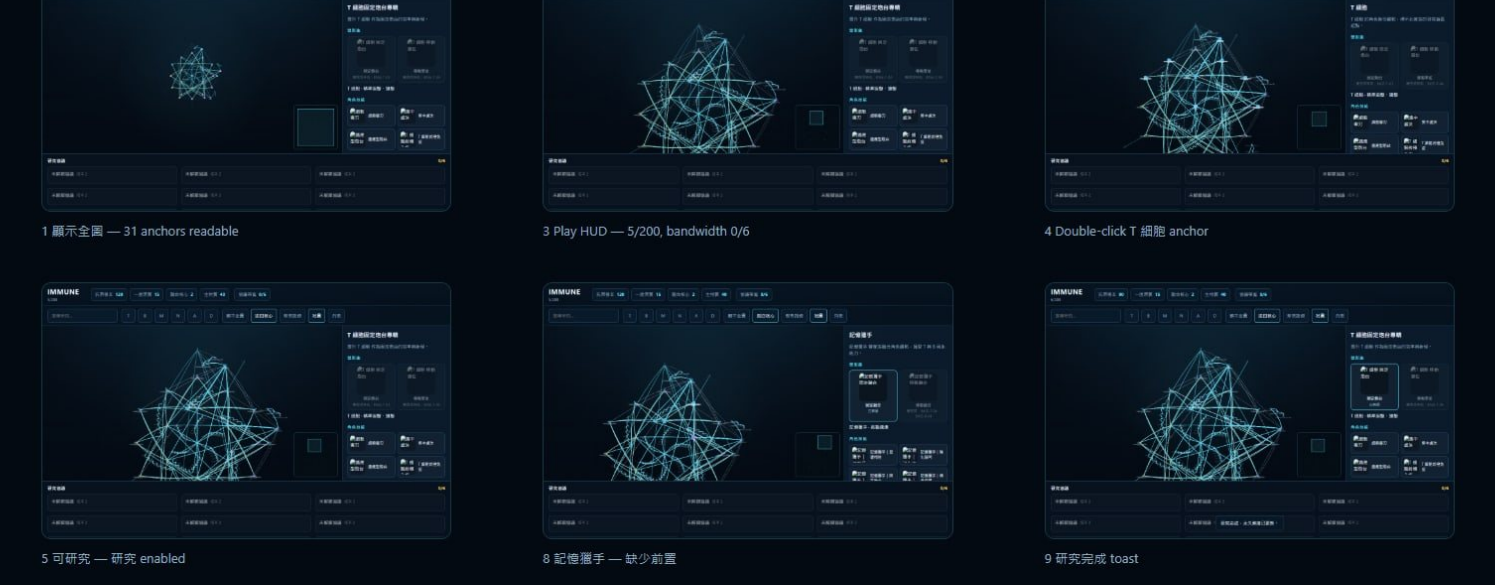
Layer	Contents
Core	免疫核心 — single origin, no cost
Inner rings	6 universal techs × 7 + 7 status chemistry
Family ring	6 bases × 8 exclusive researches + identity anchors
Pair ring	15 dual-family fusions × (S1/S2/anchor/S4 protocol)
Outer rings	6 triples + 3 hidden Apex + IMMUNE PRIME

CORE LOOP

Open star map → Pan / zoom 0.35×–2.4× → Select node → 追蹤 or 研究 → Permanent unlock → Equip 研究協議

Win: toast 「研究完成，永久進度已更新。」 and newly revealed nodes glow. Fail: 研究 stays disabled; toast uses engine codes (`missing_prerequisite`, `missing_resource`, `track_limit`, `bandwidth_exceeded`).

UX STORY — VERTICAL SLICE



Desktop 1920×1080 — detail 380px, minimap 180px, protocol row



1280×720 landscape — protocol dock peeks to 44px until hover

MECHANICS FROM ENTRY

Control	Live behaviour
Drag empty map	Pan. Wheel/pinch zooms 0.35×–2.4× around cursor.
Click node	SELECT_NODE + route emphasis. Double-click character_anchor focuses 1.05×.
顯示全圖 / 返回核心 / 聚焦路線	fitAll · home 1500,1500 @0.55× · focus selected
追蹤 / F	Max 3. Gold stroke + 追蹤中 chip. Hidden nodes cannot be tracked.
研究 / R	Only eligibility=ready. Deducts costs, completes, reveals next.
地圖 / 列表	List groups by ring; hidden names stay 未知. Minimap hidden in list.
搜尋 / family chips	Search never leaks hidden names. Filters dim nodes, never rearrange.
研究協議	Cost 1 / core 2 / Apex 3. One Apex. Bandwidth starts at 6, later 10. 裝備 / 卸下 free out of battle.
LOD	0.35–0.55 anchors only · 0.56–1.0 nodes · 1.01–1.6 names · 1.61+ costs

CATALOG — 31 IDENTITIES IN PLAY ORDER



Status chemistry (7): 標記 · 抗體 · 腐蝕 · 緩速 · 感染 · 鎮靜 · 擊暈. Universals (6×7): 防守工程 · 遠征探索 · 全面戰爭 · 細胞機動 · 融合工程 · 生存修復.

PROGRESSION / DIFFICULTY

Band	Target	Notes from spec + demo
Vertical slice	18–24 revealed, 2–4 unlocks / 15–20 min	Live demo: 22 revealed, 5 completed, antigen 120 / protomass 15 / fusion 2 / biomass 40
Main 4–6h	~45–60 completed	Family exclusives + first pair S2
Full 15–25h	~130–160 discovered	Triples + Apex gates. 200/200 is mastery, not required
Track limit	3	System shows prereqs + total cost; never auto-buys
Protocols	6 → 10 bandwidth	Normal 1 · core 2 · Apex 3 · one Apex per mission

STORE COVERS



3:4 store tile · cover-3x4.webp · 1024×1536



16:9 catalog banner · cover-16x9.webp · 1536×1024

REFRESH

Demo: [http://127.0.0.1:5180/ui/immune-research-network/](#)

21:9 board: [docs/ux-wireframe.html](#) · Manager: [docs/manager-overview.html](#)

Recapture HUD: `PORT=5180 npm run serve` then `PORT=5180 npm run docs`

Posters only: `PORT=5180 npm run docs:posters`

Validate: `bash .agents/skills/overview-pack/scripts/validate-overview-pack.sh`